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Tank 38 Solids Analyses

C. L. Crawford

October 2024

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EXECUTIVE SUMMARY

The October 2023 Tank 38 HTF-38-23-96 VDS was reported to contain 3.0 ± 0.1 wt% insoluble solids and warm acid strike dissolution indicated presence of relative high amounts of High-Level Waste (HLW) sludge components such as Al, Ca, Fe, Mn and U. Savannah River Mission Completion (SRMC) personnel thus sampled the Tank 38 in early February 2024 to provide the HTF-38-24-21 VDS for insoluble solids analysis at the Savannah River National Laboratory (SRNL). This report presents results for the elemental, radionuclide and crystalline species analyses of the February 2024 HTF-38-24-21 washed and dried solids.

The wt% insoluble solids measured for the HTF-38-24-21 VDS sample is 6.1 ± 0.7 wt%. Major elements detected in the washed and dried Tank 38 solids are Al and Na at over 10 wt%, followed by Ca, Fe, Mn, Si and U in the range of 1.6 to 8.7 wt%. Trace levels of Ba, Zn and Zr are detectable at ~ 0.02 wt% with higher levels of Cr (0.12 wt%), Ni (0.25 wt%) and Mg (0.46 wt%) present. Some elements were not detectable including B, Gd, Li and Ti. Comparison of the Tank 38 solids elemental concentrations to recent Sludge Batch 9 (SB9) dried sludge slurry indicates that the Tank 38 solids compare similarly with most sludge species except that the Na/Fe, Al/Fe and Si/Fe ratios for the Tank 38 solids are all significantly higher than those respective ratios in SB9 sludge. The SB9 sludge composition is deemed the most reasonable sludge composition for comparison to the February 2024 Tank 38 sample since no transfers have been made from Tank 22 to Tank 38 since January of 2023 and processing of SB10 did not start until March of 2023. Comparison of the typical sludge species Mn, Ni and U to Fe ratios for either the HTF-38-24-21 solids (Mn/Fe = 0.4, Ni/Fe = 0.04, U/Fe = 0.9) or the SB9 slurry (Mn/Fe = 0.33, Ni/Fe = 0.07, U/Fe = 0.2) indicate similar respective ratios suggesting that the insoluble solids in the HTF-38-24-21 sample are indeed sludge-derived solids. One exception to this observation is the higher total uranium levels (either as ratio to iron, or as wt% solids) in the Tank 38 solids vs. sludge.

Similar comparison of the Tank 38 solids elemental concentrations was made to recent SB9 Defense Waste Processing Facility (DWPF) pour stream glass elemental concentrations. Several of the Tank 38 elementals are higher than SB9 glass (Al, Ca and U). Many of the Tank 38 elementals are similar to those in glass (Fe, Mn, Na, Zn and Zr). However, the Si in Tk 38 is ~ 2.6X lower compared to SB9 glass (8.7 wt% vs. 23 wt%). Additionally, several of the glass elementals (B, Li and Ti) are not detectable in the Tk 38 solids. These data indicate that the Tank 38 solids are not consistent with the elemental signature expected from SB9 pour stream glass.

A suite of radionuclides was also analyzed for the Tank 38 solids. These radionuclides including isotopes of U and Pu and Am/Cm, as well as Cs-137 and Sr-90 are noted to be in similar relative abundance to those of SB9 sludge. One exception is the higher total uranium wt% in the Tank 38 solids relative to SB9 sludge.

Crystalline species analyses on the washed and dried Tank 38 solids indicates presence of various sodium aluminosilicate (NAS) species similar to analogous measurements performed in 2004, 2015 and 2016 on Tank 38 solids. From consideration of the elemental, radionuclide and crystalline species data presented in this report, the Tank 38 solids are judged to be most likely sodium aluminosilicate solids that also contain insoluble sludge components. Assigning all the measured Na, Al and Si to NAS components results in a calculated 64 wt% NAS and 36 wt% sludge as the estimated mass fractions. If all of the analyzed uranium in the solids is associated with the NAS, then the calculated values are 75 wt% NAS and 25 wt% sludge as the estimated mass fractions.

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LIST OF ABBREVIATIONS

CCP	Corrosion Control Program
DWPF	Defense Waste Processing Facility
ECP	Enrichment Control Program
HLW	High-Level Waste
ICP-OES	inductively coupled plasma-optical emission spectroscopy
NAS	Sodium Aluminosilicate
%RSD	percent relative standard deviation
PUTTA	Pu-thenoyltrifluoroacetone
S/ICP-MS	separation/ inductively coupled plasma-mass spectrometry
SRNL	Savannah River National Laboratory
SRMC	Savannah River Mission Completion
TTQAP	Task Technical and Quality Assurance Plan
TTR	Technical Task Request
VDS	Variable Depth Sample
XRD	X-Ray Diffraction

1.0 Introduction

This technical report provides results for the 2024 Variable Depth Sample (VDS) from H Area Tank Farm Tank HTF-38-24-21 for solids characterization. These data are needed to aid Savannah River Mission Completion (SRMC) in planning for operations of Tank 38, which is the drop tank for the 2H Evaporator and can receive Defense Waste Processing Facility (DWPF) recycle material from Tank 22. The October 2023 Tank 38 HTF-38-23-96 VDS was reported to contain 3.0 ± 0.1 wt% insoluble solids and warm acid strike dissolution indicated presence of relative high amounts of High-Level Waste (HLW) sludge components such as Al, Ca, Fe, Mn and U.[1] Per guidance of the Task Technical and Quality Assurance Plan (TTQAP) for the Enrichment Control Program (ECP),[2] any samples with higher than 1 wt% insoluble solids will be analyzed by more aggressive methods such as aqua regia or peroxide fusion dissolution of the washed insoluble solids. Savannah River Mission Completion (SRMC) personnel thus sampled the Tank 38 in early February 2024 to provide the HTF-38-24-21 VDS, which was sampled 230.5 inches from the bottom of the tank, for insoluble solids analysis at the Savannah River National Laboratory (SRNL). Initial elemental characterization of the washed and dried filtered solids from the HTF-38-24-21 VDS have previously been reported.[3] Previous core samples and VDS samples from Tank 38 have been analyzed to investigate presence of observed insoluble solids.[4,5,6,7] Feed limits have been established for the 2H-Evaporator system to ensure nuclear criticality is not possible and corrosion is minimized.[8] These limits are protected by the Enrichment Control Program (ECP) and the Corrosion Control Program (CCP) that require periodic sampling and analysis to confirm that the waste supernate composition stays within the limits.[9,10] The ECP sample characterization was requested via a Technical Task Request (TTR).[11]

2.0 Experimental

2.1 Technical Details and Procedures

Details of the wt% insoluble solids analysis on the HTF-38-24-21 VDS were presented in the initial Tank 38 solids characterization memorandum.[3] The wt% insoluble solids measured for the HTF-38-24-21 sample is 6.1 ± 0.7 wt%. These washed and dried insoluble solids were subjected to both aqua regia and peroxide fusion digestions that used nominally 0.1 g dried solids digested up to a total of 100 mL. These digested solids were analyzed by Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES) for elementals characterization. A high purity water sample was used as a reagent blank for both digestion methods. The aqua regia digestions were also submitted for various radiochemical analyses. Methods typically used in the ECP analyses[1] were applied for uranium isotopes (separation/inductively coupled plasma-mass spectroscopy (S/ICP-MS) and determination of plutonium isotopes by radiochemical separation/extraction with thenoyltrifluoroacetone and counting methods (PUTTA chemical separation coupled with alpha spectroscopy). Additionally, the digested samples were also submitted for 1) gamma spectroscopy to determine the Cs-137, 2) Sr-90 analysis by extraction and beta counting and 3) Americium/Curium (Am/Cm) radiochemical analysis by extraction and alpha spectroscopy. Small milligram portions of the washed and dried HTF-38-24-21 VDS powders were also analyzed by X-Ray Diffraction (XRD) using a contained XRD instrument to investigate the qualitative presence of crystalline species.

2.2 Quality Assurance

Requirements for performing reviews of technical reports and the extent of review are established in Manual E7, Procedure 2.60.[12] SRNL documents the extent of the Design Check using the SRNL Technical Report Design Checklist.[13] This review meets the acceptable criteria to comply with the TTR[2] requesting this work

with a functional classification of Safety Class and per guidance in the TTQAP.[11] Data are recorded in the electronic laboratory notebook system as notebook/experiment number B9108-00327-41.[14]

3.0 Results and Discussion

Elemental Analyses

Table 3-1 provides the average Tank 38 washed and dried solids chemical analyses for the combined triplicate samples from both aqua regia and peroxide fusion. Estimated one sigma percent uncertainty values that provide an indication of the uncertainty associated with the ICP-OES are 5%. Some of the components only use aqua regia (B, Ba, Gd, Li, Na, Ti, Zn and Zr) and Si is determined by peroxide fusion only. Deionized water blanks showed no detectable species except for Ca (0.3 wt%) and Mg (0.1 wt%) in aqua regia and Ca (0.8 wt%) in peroxide fusion. Major metals shown are Al and Na at over 10 wt%, followed by Ca, Fe, Mn, Si and U in the range of 1.6 to 8.7 wt%. Trace levels of Ba, Zn and Zr are detectable at ~ 0.02 wt% with higher levels of Cr (0.12 wt%), Ni (0.25 wt%) and Mg (0.46 wt%) present. Some elements are listed as not detectable including B, Gd, Li and Ti. The combined average of Sludge Batch 9 (SB9) dried slurry elemental analyses[15] are shown in the next to last column of Table 3-1 for comparison to the HTF-38-24-21 solids. There have been no transfers from Tank 22 to Tank 38 since January of 2023 and processing of SB10 sludge did not start until March of 2023, so the SB9 sludge composition should be the most reasonable comparison. The highest elemental components of SB9 sludge slurry are Fe (16 wt%) and Na (14.7 wt%) at over 10 wt%, with Al, Mn, Ni, Si and U in the 1.1 to 6.3 wt% range.

Comparison of the individual components to the Fe content for either HTF-38-24-21 or SB9 slurry indicates that the Al, Na and Si are elevated in the Tank 38 solids vs. their respective ratios to Fe in the SB9 slurry. The Al/Fe, Na/Fe and Si/Fe ratios in the Tank 38 solids are 1.7, 1.8 and 1.4, respectively. This same comparison for SB9 sludge slurry shows lower Al/Fe, Na/Fe and Si/Fe ratios of 0.4, 0.9 and 0.1, respectively. One explanation for the increased levels of Na, Al and Si in the Tank 38 solids vs. sludge could be the presence of sodium aluminosilicate solids. These crystalline species have previously been identified in Tank 38 solids analyses reported in 2004.[4] This observation is also supported by the current XRD analyses of the HTF-38-24-21 solids sample as discussed below in the XRD analyses section. Comparison of the typical sludge species Mn, Ni and U to Fe ratios for either the HTF-38-24-21 solids (Mn/Fe = 0.4, Ni/Fe = 0.04, U/Fe = 0.9) or the SB9 slurry (Mn/Fe = 0.33, Ni/Fe = 0.07, U/Fe = 0.2) indicate similar respective ratios suggesting that the insoluble solids in the HTF-38-24-21 sample are indeed sludge-derived solids. One exception to this observation is the higher total uranium levels (either as ratio to iron, or as wt% solids) in the Tank 38 solids vs. sludge. The HTF-38-24-21 solids analyses shown in Table 3-1 can also be compared to the elemental analyses from recent SB9 DWPF pour stream glass.[16] The elemental composition of the SB9 glass is shown in the last column of Table 3-1. Several of the Tank 38 elementals are higher than SB9 glass (Al, Ca and U). Many of the Tank 38 elementals are similar to those in glass (Fe, Mn, Na, Zn and Zr). However, the Si in Tk 38 is ~ 2.6X lower compared to SB9 glass (8.7 wt% vs. 23 wt%). Additionally, several of the glass elementals (B, Li and Ti) are not detectable in the Tk 38 solids. These data indicate that the Tank 38 solids are not consistent with the elemental signature expected from SB9 pour stream glass or any other DWPF glasses.

The measured elemental composition of the HTF-38-24-21 sample can also be compared to previous Tank 38 solids analyses.[6,7] If one assumes that all of the Na, Al and Si in these various solids are associated with sodium aluminosilicate phases, then the fraction of NAS to total measured elemental solids in these samples is estimated to be 76 wt% NAS,[6] 67 wt% NAS,[7] and 64 wt% NAS for the current sample. The remaining elemental solids in these samples to make up 100 wt% are sludge-containing components such as Ca, Fe,

Mn, Ni and U. It is known that uranium can be associated with the NAS solids via sorption, co-precipitation and occlusion.[17] If uranium is included as part of the NAS calculations, results above become estimated 76 wt% NAS,[6] 80 wt% NAS,[7] and 75 wt% NAS for the current sample.

Table 3-1. Washed and Dried Solids Analyses for Tank 38

	HTF-38-24-21 Washed/Dried Solids			SB9 Slurry*	SB9 Glass**
Element	Avg. (wt%)	St. Dev. (wt%)	%RSD	Avg. (wt%)	Avg. (wt%)
Al	10.6	1.8	16.7	6.34	3.4
B	<0.01	--	--	0.03	1.5
Ba	0.020	0.001	5.3	0.07	0.03
Ca	1.6	0.4	27.4	0.98	0.50
Cd	0.031	0.003	10.8	0.01	0.01
Cr	0.12	0.014	11.5	0.09	0.05
Fe	6.2	0.63	10.1	16.03	7.7
Gd	<0.01	--	--	0.08	0.05
Li	<0.33	--	--	0.04	2.0
Mg	0.46	0.042	9.2	0.22	0.10
Mn	2.7	0.3	10.1	5.22	2.0
Na	11.0	0.80	7.3	14.73	10.6
Ni	0.25	0.02	10.0	1.10	0.53
Si	8.7	0.1	1.6	1.22	23.0
Ti	<0.02	--	--	0.02	1.4
U	5.4	0.4	8.1	3.0	1.4
Zn	0.017	0.002	11.2	0.03	0.015
Zr	0.018	0.001	3.7	0.06	0.034

*See 'Combined Average' column of Table 3-6 of Reference 15 for dried sludge slurry elemental concentrations.
**Elemental wt% composition determined from glass oxide species from Table 3-3 in Reference 16.

Radionuclide Analyses

Table 3-2 provides average Tank 38 washed and dried solids radiochemical analyses for the triplicate samples from aqua regia. All radionuclides reported for the U, Pu, Cs-137 and Sr-90 isotopes were detectable. Am/Cm analyses indicate Am-241, -243 and -242m and Cm-242, -244 are detectable. The U-235/Total U ratio for the Tk 38 solids is 0.65% and the total uranium is 5 wt% in the washed and dried solids as calculated from the S/ICP-MS data. This total uranium number is in reasonable agreement with the total uranium value of 5.4 wt% determined from the ICP-OES analysis. Pu-238 is the highest of the various Pu isotopes concentration by mass. All radionuclide analyses for the tank 38 solids, with the exception of uranium, indicate that these radionuclides are in similar relative abundance to those reported for sludge batch 9 radionuclide analysis.[18] Comparable concentrations for Cs-137, Sr-90, Am-241 and Am-243 were also reported from the 2015 Tank 38 solids analyses as shown in the note at the bottom of Table 3-2.

Estimated one sigma percent uncertainty values that provide an indication of the uncertainty associated with the various radiochemical counting methods are 5% and 10% for Cs-137 and U isotopes, respectively. One-sigma uncertainty values for Pu-241 are 15.1% to 16.1% and are 5.5% to 7.8% for both Pu-238 and Pu-239/240. The one-sigma uncertainty values are as follows for the Am and Cm isotopes: Am-141 (11.1%

to 11.8%), Am-243 (11.3% to 12.0%) and Am-242m (18.5% to 38.9%), Cm-242 (18.5% to 38.9%) and Cm-244 (11.1% to 11.8%).

Table 3-2. Washed and Dried Solids Radionuclide Analyses for Tank 38

analyte	method	units	HTF38-24-21	
			average	%RSD
Sample Type	--	--	Subsurface Solids	
U-233	S/ICPMS	wt%	2.93E-04	7.4%
U-234	S/ICPMS	wt%	8.36E-03	6.2%
U-235	S/ICPMS	wt%	3.23E-02	7.6%
U-236	S/ICPMS	wt%	3.97E-03	6.4%
U-238	S/ICPMS	wt%	4.92E+00	7.3%
Total U	calc.	wt%	4.96E+00	7.3%
U-235 / U	calc.	%	0.65%	0.4%
Pu-238	PuTTA	wt%	2.83E-01	1.6%
		dpm/g	3.91E+08	
Pu-239 [a]	PuTTA	wt%	5.09E-06	1.2%
Pu-239/240	PuTTA	dpm/g	1.16E+07	
Pu-241	Pu238/41	wt%	2.45E-04	20%
		dpm/g	9.30E+07	
Cs-137	Gam.Scan	dpm/g	4.42E+08	4.9%
Ba-137m	calc.		4.18E+08	
Sr-90	Sr-90	dpm/g	2.15E+09	6.7%
Y-90	calc.		2.15E+09	
Am-241	Am/Cm	dpm/g	1.64E+07	4.4%
Am-243	Am/Cm	dpm/g	7.02E+05	5.5%
Am-242m	Am/Cm	dpm/g	3.58E+04	33.0%
Cm-243	Am/Cm	dpm/g	<4.75E+05	NA
Cm-245	Am/Cm	dpm/g	<6.59E+04	NA
Cm-247	Am/Cm	dpm/g	<2.09E+04	NA
Cf-249	Am/Cm	dpm/g	<2.14E+04	NA
Cf-251	Am/Cm	dpm/g	<1.88E+04	NA
Cm-242	Am/Cm	dpm/g	2.96E+04	33.1%
Cm-244	Am/Cm	dpm/g	2.97E+07	7.4%

calc. = calculation; [a]Pu-239 mass assumes entire Pu-239/240 activity is Pu-239

Note: Similar concentrations were reported in 2015[6] for the following radionuclides from dried, unwashed Tank HTF-38-15-1 solids in units of dpm/g: Cs-137 = 3.45E+08, Sr-90 = 4.80E+09, Am-241 = 1.03E+07, Am-243 = 5.68E+05.

Crystalline Species Analyses

Milligram quantities of the Tk 38 solid powders were analyzed for crystalline species for 3 separate triplicate aliquots obtained from the parent sample jar using XRD and data collection times of ~ 3 hrs. The results of the XRD analyses on the three separate powder samples are shown in Figure 3-1, Figure 3-2 and Figure 3-3. The XRD patterns are similar for all three samples that show a broad amorphous 'hump' centered around the two-theta degrees value of ~ 25. This broad amorphous hump is typically seen for High Level Waste (HLW) amorphous silicate glass powder samples.[19]

All patterns contain similar magnitude peaks indicating various crystalline species present. The [00-054-0628] sodium aluminosilicate cancrinite containing nitrate, the hematite and the sodium aluminosilicate natrolite are common to all three samples. Samples 2 and 3 also contain [00-55-1018] sodium aluminosilicates, and uranium oxide hydrate. Sample 1 also contains crystalline patterns for $NaNO_3$ and SiO_2. Various sodium aluminosilicates as well as hematite crystalline materials have also been identified in the solid core samples previously analyzed from Tank 22,[20] which is the DWPF recycle tank that feeds into Tank 38. Sodium aluminosilicates including hydroxycancrinite [31-1272] and Zeolite Y [434-0168] have also been identified in a 2004 study of Tank 38 solids.[4] The sodium aluminosilicate pattern identified for these Tank 38 solids as the 00-054-0628 species is very similar to the crystalline pattern for the 31-1272 hydroxycancrinite sodium aluminosilicate identified in the 2004 study. Sodium aluminosilicate crystalline phases were also identified as the peaks with highest intensity in the previous 2015 and 2016 analyses.[6,7]

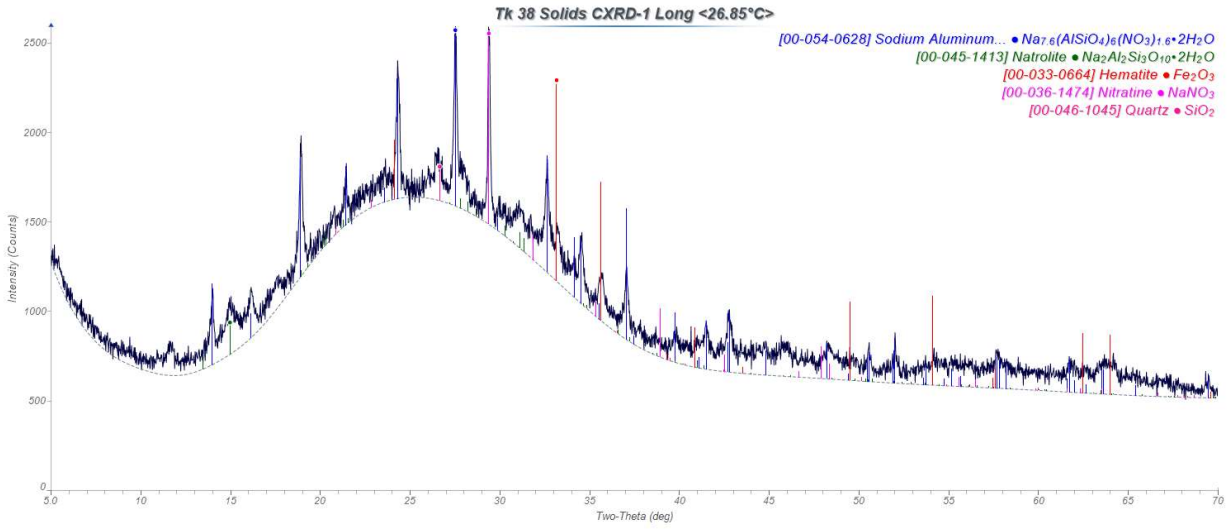


Figure 3-1. X-ray Diffraction Pattern for Tank 38 Solids Sample 1 of 3

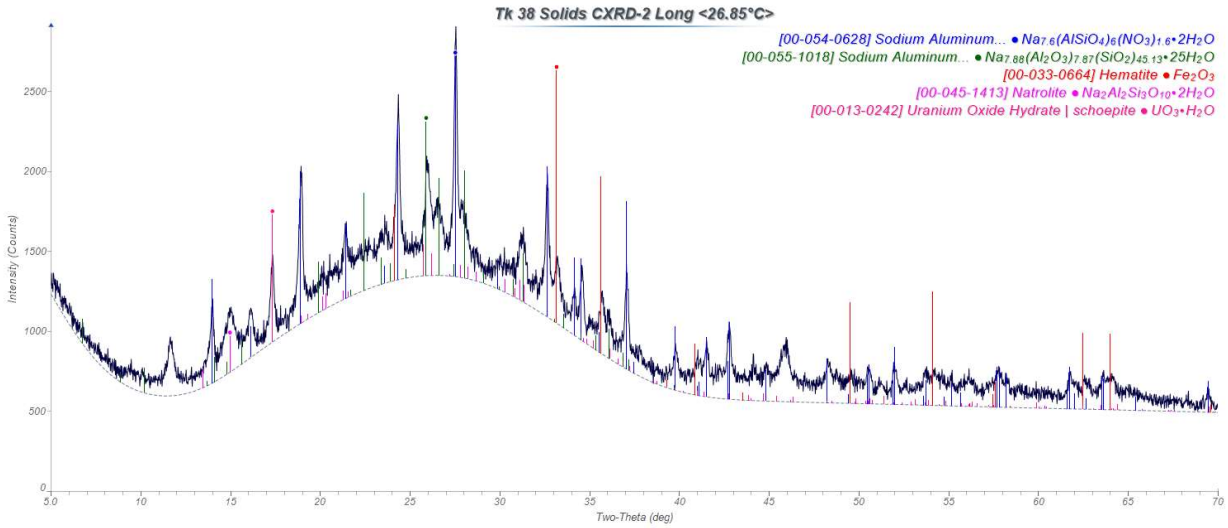


Figure 3-2. X-ray Diffraction Pattern for Tank 38 Solids Sample 2 of 3

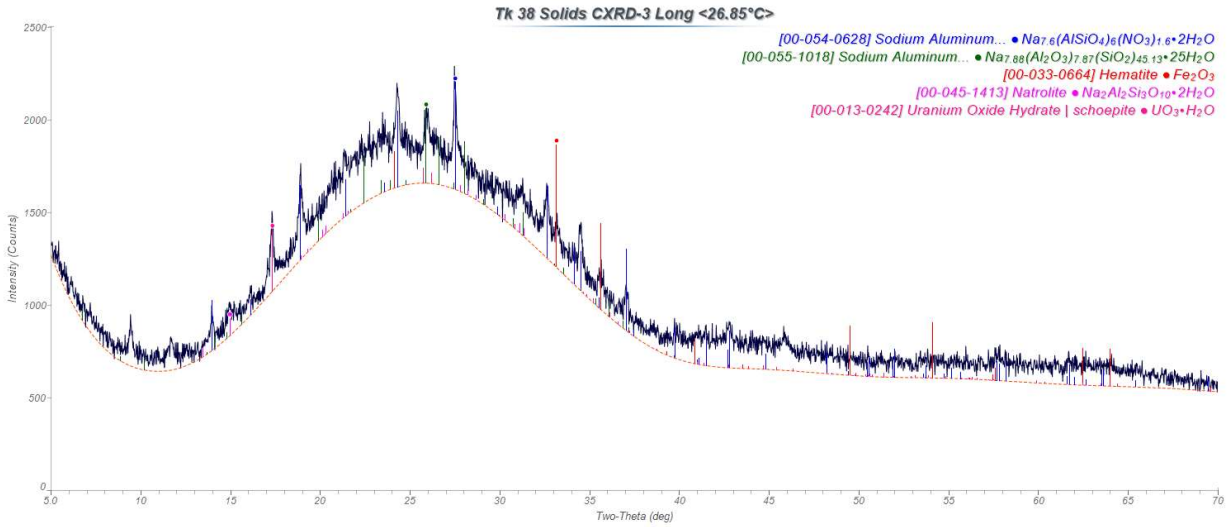


Figure 3-3. X-ray Diffraction Pattern for Tank 38 Solids Sample 3 of 3

4.0 Conclusions

The wt% insoluble solids measured for the HTF-38-24-21 VDS sample is 6.1 ± 0.7 wt%. Major elements detected in the washed and dried Tank 38 solids are Al and Na at over 10 wt%, followed by Ca, Fe, Mn, Si and U in the range of 1.6 to 8.7 wt%. Trace levels of Ba, Zn and Zr are detectable at ~ 0.02 wt% with higher levels of Cr (0.12 wt%), Ni (0.25 wt%) and Mg (0.46 wt%) present. Some elements were not detectable including B, Gd, Li and Ti. Comparison of the Tank 38 solids elemental concentrations to recent Sludge Batch 9 (SB9) dried sludge slurry indicates that the Tank 38 solids compare favorably with most sludge species except that the Na/Fe, Al/Fe and Si/Fe ratios for the Tank 38 solids are all significantly higher than those respective ratios in SB9 sludge. The SB9 sludge composition is deemed the most reasonable sludge composition for comparison to the February 2024 Tank 38 sample since no transfers have been made from Tank 22 to Tank 38 since January of 2023 and processing of SB10 did not start until March of 2023. Comparison of the typical sludge species Mn, Ni and U to Fe ratios for either the HTF-38-24-21 solids (Mn/Fe = 0.4, Ni/Fe = 0.04, U/Fe = 0.9) or the SB9 slurry (Mn/Fe = 0.33, Ni/Fe = 0.07, U/Fe = 0.2) indicate similar respective ratios suggesting that the insoluble solids in the HTF-38-24-21 sample are indeed sludge-derived solids. One exception to this observation is the higher total uranium levels (either as ratio to iron, or as wt% solids) in the Tank 38 solids vs. sludge.

Similar comparison of the Tank 38 solids elemental concentrations was made to recent SB9 Defense Waste Processing Facility (DWPF) pour stream glass elemental concentrations. Several of the Tank 38 elementals are higher than SB9 glass (Al, Ca and U). Many of the Tank 38 elementals are similar to those in glass (Fe, Mn, Na, Zn and Zr). However, the Si in Tk 38 is ~ 2.6X lower compared to SB9 glass (8.7 wt% vs. 23 wt%). Additionally, several of the glass elementals (B, Li and Ti) are not detectable in the Tk 38 solids. These data indicate that the Tank 38 solids are not consistent with the elemental signature expected from SB9 pour stream glass.

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Crystalline species analyses on the washed and dried Tank 38 solids indicates presence of various sodium aluminosilicate (NAS) species similar to analogous measurements performed in 2004, 2015 and 2016 on Tank 38 solids. From consideration of the elemental, radionuclide and crystalline species data presented in this report, the Tank 38 solids are judged to be most likely sodium aluminosilicate solids that also contain insoluble sludge components. Assigning all of the measured Na, Al and Si to NAS components results in a calculated 64 wt% NAS and 36 wt% sludge. If all the uranium is associated with the NAS solids, the calculated values are 75 wt% NAS and 25 wt% sludge.

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